Retail Sales Analysis Project Report

Python data preparation, exploratory findings, MySQL storage, and Power BI dashboard

1. Project overview

This project analyzes 10,194 retail sales-line records to understand revenue, profit, product mix, regional performance, customer segments, discounts, and time-based patterns. Python was used to validate and summarize the data, the prepared records were loaded into MySQL through XAMPP, and Power BI was used to build an interactive dashboard.

Business problem

Retail decision-makers need to see where sales convert into profit and where performance varies by product, region, customer segment, and time. High sales alone do not guarantee strong profit, so the data should be reviewed using both revenue and profitability measures.

How can retail sales data reveal performance trends and areas for investigation so the business can improve profitability and make better product, customer-segment, and regional decisions?

KPI snapshot

Total sales	Total profit	Overall profit margin	Units sold
$2,326,534.35	**$292,296.81**	**12.56%**	**38,654**

Dataset summary

The table contains 21 fields: order and shipping dates, ship mode, customer and order identifiers, customer segment, geographic fields, product category and sub-category, sales, quantity, discount, and profit. The data spans January 3, 2023 through December 30, 2026. Customer age, gender, subscription, review, and sales-channel attributes are not present, so this report focuses on transaction and segment behavior rather than demographic behavior.

Project workflow

Python for validation and exploratory analysis | **MySQL / XAMPP** for structured storage | **Power BI** for interactive reporting

2. Python data preparation and quality checks

The notebook outputs show a structured review using the DataFrame dimensions, column types, descriptive statistics, totals, and grouped summaries. The retail file was independently checked against the displayed Python results before the report figures were prepared.

Check	Result
Dataset size	10,194 records and 21 columns.
Missing values	No missing values found across the supplied fields.
Duplicate records	No fully duplicated rows; Row ID is unique.
Date fields	Order Date and Ship Date are parsed as dates.
Numeric fields	Sales, Profit, and Discount are decimal values; Quantity and Row ID are whole numbers.
Category and ID fields	Names, product/customer/order IDs, location, segment, and category fields are treated as text.

Descriptive statistics

Measure	Mean	Median	Minimum	Maximum
Sales per line	$228.23	$53.91	$0.44	$22,638.48
Profit per line	$28.67	$8.69	-$6,599.98	$8,399.98
Quantity per line	3.79	3	1	14
Discount rate	15.54%	20%	0%	80%

Interpretation

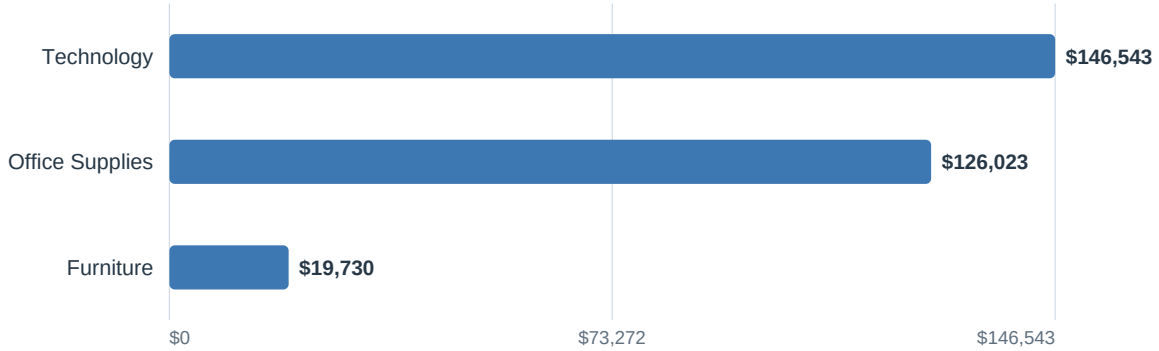
- Average sales per line ($228.23) are much higher than the median ($53.91), indicating that a relatively small number of high-value lines pull the average upward.
- Profit is widely spread: the lowest line is a loss of about $6,600, while the highest line earns about $8,400. Review these large values against the original transactions before classifying them as errors.
- The mean discount is 15.54%, while the median is 20%. This is a reason to test how discounts relate to profit by category and sub-category, not proof that discounts caused a result.

3. Key findings from Python

Total sales are **$2,326,534.35** and total profit is **$292,296.81**, producing an overall profit margin of **12.56%** (total profit divided by total sales). The dataset records 38,654 units sold. The results below compare total profit and average profit per sales line.

Profit by product category

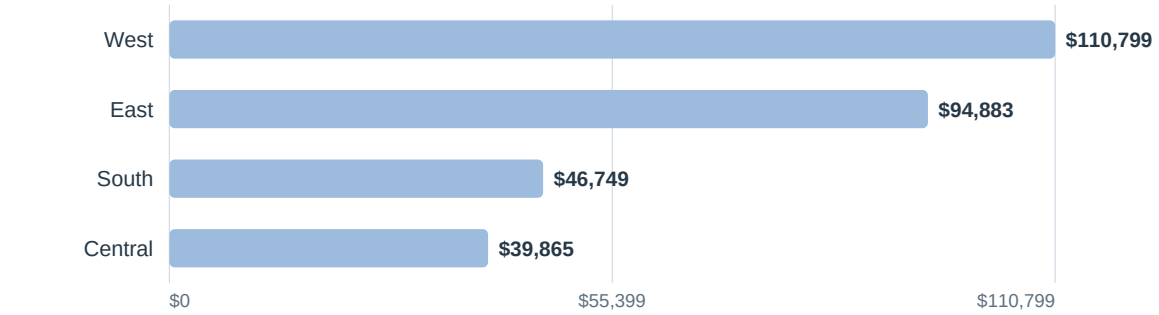
Category	Sales lines	Total profit	Average profit per line	Share of total profit
Technology	1,865	$146,543.38	$78.58	50.1%
Office Supplies	6,128	$126,023.44	$20.57	43.1%
Furniture	2,201	$19,730.00	$8.96	6.7%



Technology is the largest profit contributor at $146,543.38, about half of total profit, and has the highest average profit per line. Office Supplies has the most sales lines and contributes $126,023.44. Furniture contributes $19,730.00 and has the lowest average profit per line. These totals identify where to investigate product mix, discounting, and loss-making sub-categories.

Profit by region

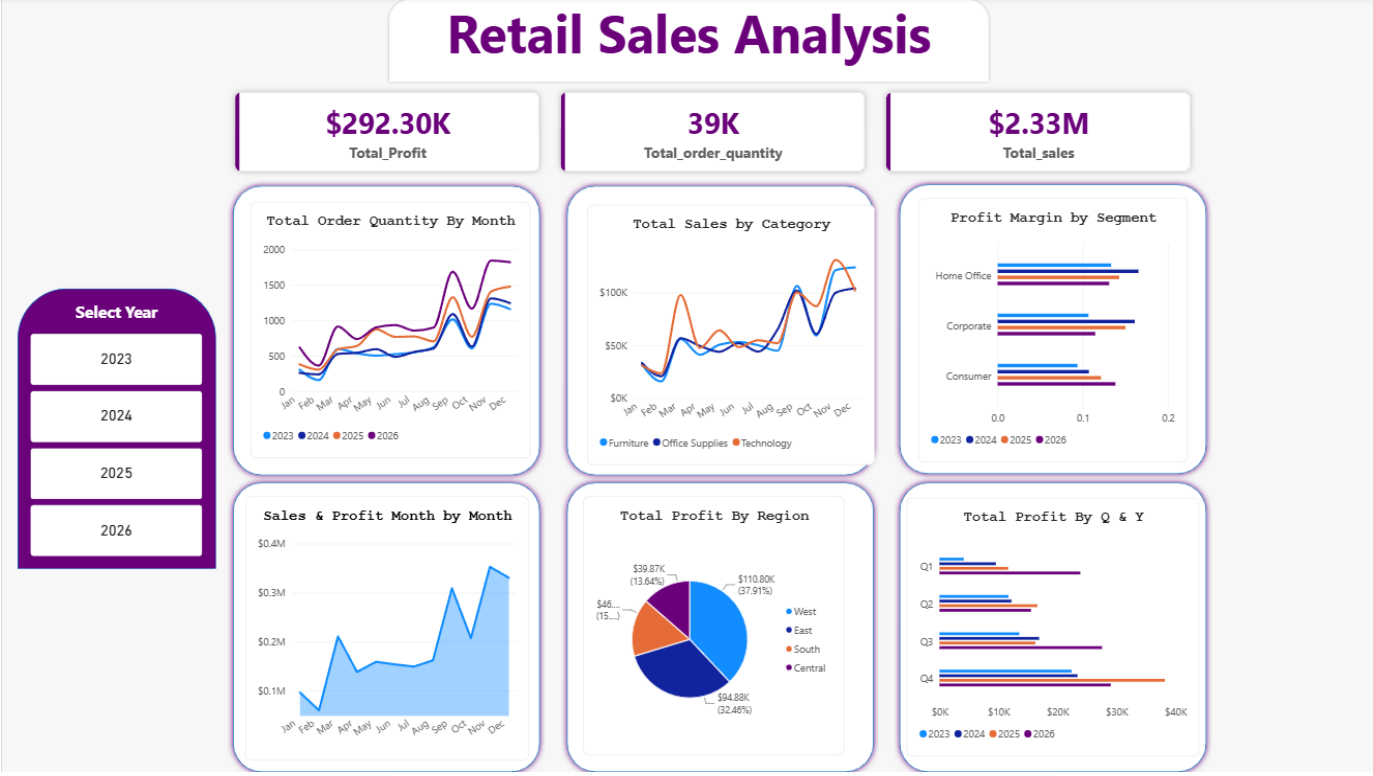
Region	Sales lines	Total profit	Average profit per line	Share of total profit
West	3,253	$110,798.82	$34.06	37.9%
East	2,986	$94,883.26	$31.78	32.5%
South	1,620	$46,749.43	$28.86	16.0%
Central	2,335	$39,865.31	$17.07	13.6%



The West contributes the most profit ($110,798.82), followed by the East ($94,883.26). Together they account for about 70.4% of total profit. Central has the lowest total profit ($39,865.31) and lowest average profit per line ($17.07). These figures are profit totals, not regional profit margins.

4. Power BI dashboard

The dashboard combines headline KPIs with a year selector and visuals for monthly quantity, category sales, segment profit margin, monthly sales, regional profit, and quarterly profit by year. This allows performance to be reviewed by year and across products, customer segments, regions, and time periods.



Dashboard screenshot supplied for this report. The KPI cards display rounded totals of $2.33M sales, $292.30K profit, and 39K units; the Python calculations provide the exact values used in this report.

Business questions answered by the dashboard

- What are total sales, total profit, and total order quantity for the selected year?
- How does total order quantity change month by month across the selected years?
- How do monthly sales trends differ among Furniture, Office Supplies, and Technology?
- How does profit margin differ among Consumer, Corporate, and Home Office segments by year?
- How do monthly sales change during the year, and which months show the highest or lowest sales?
- How is total profit distributed among the West, East, South, and Central regions?
- How does total profit vary by quarter and year, and which quarters show the highest or lowest profit?

Visual check: The visual titled “Sales & Profit Month by Month” appears to display the sales series only. Add Total Profit to the visual if it should compare both measures, or rename it to “Total Sales by Month.”

5. Business recommendations and next steps

Use the findings to guide follow-up analysis and targeted decisions. The figures support areas to investigate; they do not by themselves establish the cause of a profit difference.

Evidence	Recommended action
Technology contributes about 50.1% of total profit and has the highest average profit per line.	Protect availability of its strongest products and identify which products or sub-categories account for the profit contribution.
Office Supplies has 6,128 sales lines and contributes about 43.1% of total profit.	Review high-volume, low-profit sub-categories and discount levels to find opportunities to protect profit per line.
Furniture contributes about 6.7% of total profit and has the lowest average profit per line.	Investigate product and sub-category results, discounts, and loss-making lines before changing assortment or promotions.
West and East together generate about 70.4% of total profit; Central is lowest at about 13.6%.	Retain performance in the West and East, and compare Central's product mix and discount patterns with stronger regions.
Discounts average 15.54%, with values up to 80%; profit ranges from -$6,599.98 to $8,399.98 per line.	Check extreme records and test discount-profit patterns at product level before making pricing decisions.

Interpretation boundaries

- The dataset supports analysis by customer segment, but it has no age, gender, subscription, review, or online/offline channel fields. Avoid making claims about those factors.
- Category and regional comparisons use total profit and average profit per sales line. They are not the same as profit margin by category or region.
- The Power BI monthly and quarterly visuals support trend exploration. Report exact seasonal or year-over-year changes only after checking the selected years and complete date periods.
- The MySQL/XAMPP database is part of the workflow. Confirm the Power BI connection and refresh mode before describing the dashboard as live-connected.

Conclusion

The Python analysis validates a 10,194-row retail dataset with no missing or duplicate rows and reports $2.33M in sales, $292.30K in profit, and a 12.56% overall margin. Technology and the West are the leading profit contributors in the supplied summaries, while Furniture and Central are the clearest areas for follow-up. The Power BI dashboard makes these dimensions explorable; product-level and discount analysis can now turn the descriptive findings into specific business actions.